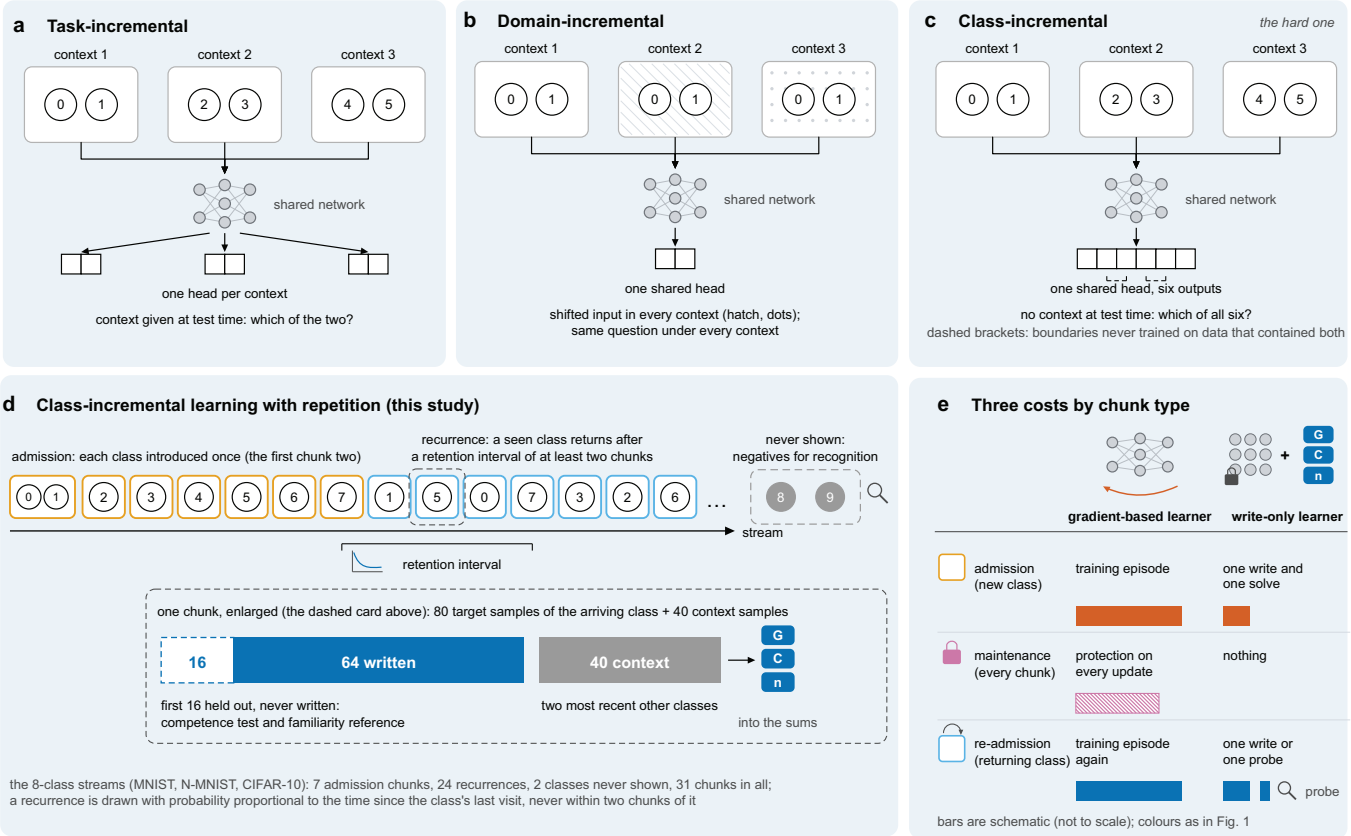

a
Task-incremental
context 1
context 2
context 3
0
1
2
3
4
5
shared network
one head per context
context given at test time: which of the two?
b
Domain-incremental
context 1
context 2
context 3
shared network
one shared head
shifted input in every context (hatch, dots);
same question under every context
c
Class-incremental
the hard one
context 1
context 2
context 3
shared network
one shared head, six outputs
no context at test time: which of all six?
dashed brackets: boundaries never trained on data that contained both
d
Class-incremental learning with repetition (this study)
admission: each class introduced once (the first chunk two)
recurrence: a seen class returns after
a retention interval of at least two chunks
never shown:
negatives for recognition
stream
retention interval
one chunk, enlarged (the dashed card above): 80 target samples of the arriving class + 40 context samples
16
64 written
40 context
G
C
n
first 16 held out, never written:
competence test and familiarity reference
two most recent other classes
into the sums
the 8-class streams (MNIST, N-MNIST, CIFAR-10): 7 admission chunks, 24 recurrences, 2 classes never shown, 31 chunks in all;
a recurrence is drawn with probability proportional to the time since the class's last visit, never within two chunks of it
e
Three costs by chunk type
gradient-based learner
write-only learner
admission
(new class)
training episode
one write and
one solve
maintenance
(every chunk)
protection on
every update
nothing
re-admission
(returning class)
training episode
again
one write or
one probe
probe
bars are schematic (not to scale); colours as in Fig. 1